A Scalable Framework for Reservoir Water-Level Monitoring Using Optical–SAR Integration in Data-Sparse Regions

Kwanghee Han, Seokhyeon Kim, Rajeshwar Mehrotra, and Ashish Sharma

***Abstract*—Monitoring water levels in lakes and reservoirs is critical for sustainable water resource management. While ground-based sensors have traditionally been used, remote sensing technologies offer a vital alternative, particularly in regions where direct measurements are costly, time-consuming, or impractical. Our objective is to develop and rigorously evaluate a scalable framework for reservoir water-level monitoring in data-sparse regions by integrating optical and synthetic aperture radar (SAR) imagery with digital elevation model (DEM)-driven, per-scene dynamic thresholding. The approach is grounded in reflectance and backscattering properties. To overcome the limitations of conventional fixed-threshold methods, which are highly time- and location-dependent and often unstable under varying hydrological conditions, our framework employs DEM-driven, per-scene dynamic thresholding to enhance accuracy, particularly for small and complex inland water bodies. Moreover, the study demonstrates that water pixels can be accurately identified using individual spectral bands, eliminating the need for conventional indices. The proposed methodology was tested using Harmonized Landsat Sentinel-2 (HLS) and Sentinel-1 datasets across four South Korean reservoirs (Chungju, Andong, Yongdam, and Juam) with diverse hydrological characteristics, showing consistently high accuracy (HLS: $R^2 \approx 1.0$ and ubRMSE = 0.37 ~ 0.65 m; Sentinel-1: $R^2 \geq 0.95$ and ubRMSE = 0.47 ~ 1.29 m). The integration of optical and SAR significantly improves water-level monitoring, achieving a high-density time series with an average observation interval of ~3 days, compared to 10–20 days for HLS and 6–10 days for Sentinel-1. The results highlight that the method demonstrates high-frequency monitoring with potential for near-real-time reservoir applications, offering new capabilities for hydrological assessments and water resources management.**

***Index Terms*—Digital elevation model (DEM)-based water mask, frequent water-level retrieval, optical–synthetic aperture radar (SAR) integration, transboundary water systems, water-level monitoring via remote sensing.**

I. Introduction

A DETAILED and accurate acquisition of hydrological information is crucial for sustainable water resource management and for addressing water conflicts. Accurate reservoir water levels are particularly important when dam operating rules are undisclosed, as operations strongly affect downstream water quantity and quality [1], [2]. Notably, approximately 50% of the world's population reside downstream of dams [3], highlighting the need to monitor dam dynamics for flood risk mitigation and equitable allocation. This importance is further amplified under climate change, which increases the frequency of floods and droughts and elevates the role of reservoirs in both routine operation and emergency response. In transboundary regions, where hydrological data are often withheld, the lack of reliable monitoring can aggravate tensions, emphasizing the value of scalable satellite-based approaches for timely water-level estimation in support of crisis management and cooperative water governance [4], [5], [6].

Previous studies have addressed water-level monitoring in unmeasured or undisclosed regions by using altimetry satellites, originally designed for oceanic sea-level measurements but now widely applied to inland water bodies like rivers, lakes, and reservoirs with high accuracy [7], [8]. Their success has promoted their adoption in hydrological monitoring. However, sparse revisit intervals, typically ranging from several weeks to months, pose challenges for capturing rapid water level changes in dynamic systems, as noted by Calmant et al. [9]. To overcome temporal gaps, researchers have explored the integration of multiple satellites to produce denser time series [5], [10]. Still, these approaches remain limited in resolving the fast and large fluctuations typical of reservoirs, highlighting the need for more frequent and precise monitoring.

Consequently, alternative satellite sensors beyond altimetry have been explored for monitoring water levels in regions with limited or sparse data. Microwave radiometer missions, such as Soil Moisture and Ocean Salinity (SMOS) and Soil Moisture Active Passive (SMAP), have been employed to estimate surface water dynamics [11], [12]. However, their coarse spatial resolutions—approximately 40 km for SMOS and 36 km for SMAP—limit their applicability [13], [14], particularly in small and complex basins where finer spatial details are important. This limitation is illustrated in Fig. 1,

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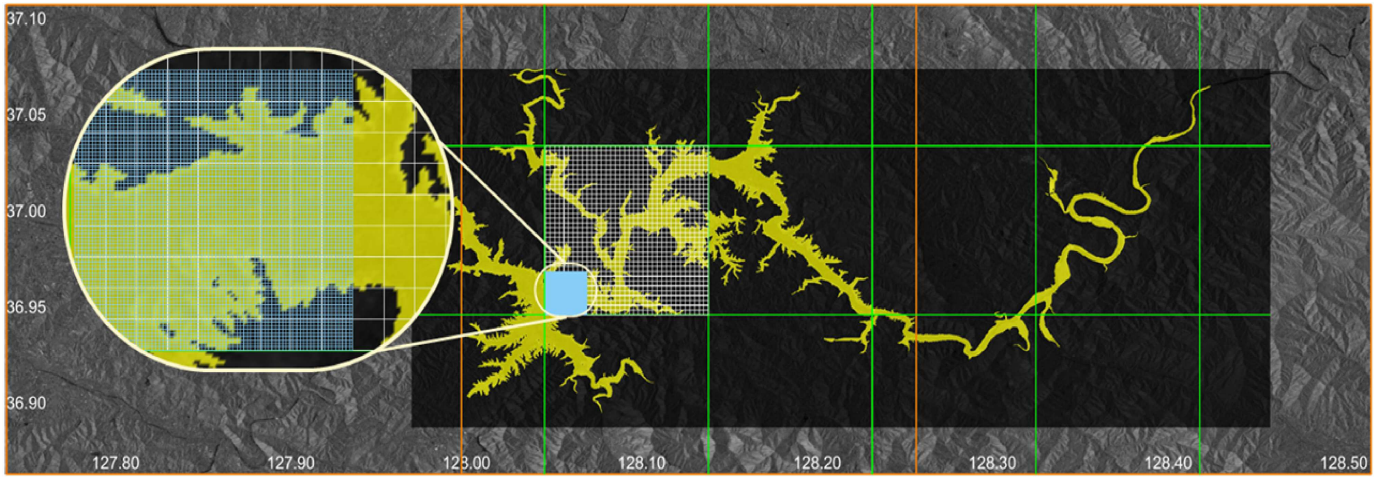


Fig. 1. Spatial resolution of various satellite datasets over Chungju Reservoir, South Korea's largest reservoir (shown for illustration). The grid represents different satellite spatial resolutions. SMOS (orange, 25 km) and SMAP (green, 9 km) provide relatively coarse spatial resolution, while MODIS (white, 250 m) and harmonized Landsat Sentinel-2 (blue, 30 m) offer finer, denser grid data. The background map data are from Sentinel-1 SAR with a 10-m resolution.

which highlights the need for higher resolution sensors in such contexts. In contrast, optical sensors and synthetic aperture radar (SAR) systems offer substantially higher spatial resolutions, often down to 1 km or even finer. These sensors provide valuable multispectral imagery for surface water mapping [15], [16], [17].

Moderate Resolution Imaging Spectroradiometer (MODIS) has been one of the most commonly used optical satellites for deriving water time series due to its extensive coverage, short observation intervals, and availability of over 20 years of free multispectral data across 36 bands [18]. However, its coarse spatial resolution (from 250 m to 1 km) limits its effectiveness in capturing dynamic hydrological changes in small and complex water bodies, although it performs adequately for large and simple ones [19]. In contrast, medium-resolution optical missions such as Landsat and Sentinel-2 offer open-access and finer detail. Specifically, the Harmonized Landsat Sentinel-2 (HLS) data products are particularly advantageous, offering a spatial resolution of 30 m [18]. In parallel, the availability of elevation information—spanning both surface digital elevation models (DEMs) and bathymetric datasets—has markedly expanded, particularly through recent advances in satellite imagery and airborne LiDAR [20], [21], [22]. In this study, we employed the LiDAR-derived DEM, which provides superior vertical accuracy compared to global products and is critical for resolving fine-scale reservoir shorelines. Together with the 30-m spatial resolution of HLS, this allows precise detection of water surface changes in small and morphologically complex water bodies.

The fundamental principle for distinguishing water and land in optical imagery relies on the lower reflectance of water surfaces compared to land. Based on this, traditional water extraction methods have applied fixed thresholds for classification not only when using a single band but also when using indices developed for water detection [23], [24], [25], [26]. However, these threshold values vary considerably across regions and time periods. Even when different satellites observe the same location simultaneously, discrepancies

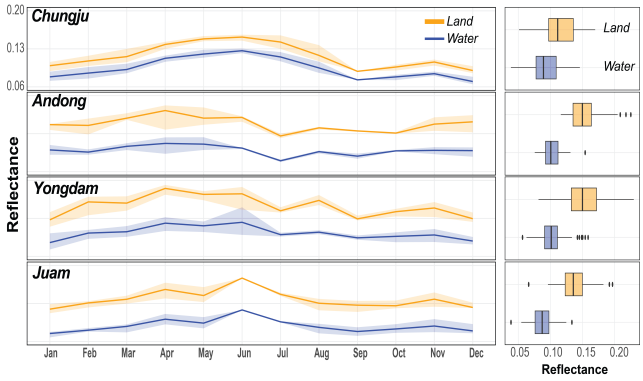


Fig. 2. Temporal and spatial variations in reflectance at the water–land boundary, derived from HLS data across four reservoirs. (Left) Monthly mean values for each reservoir with shaded areas representing the interquartile range (25th–75th percentile). (Right) Box plots summarizing the overall reflectance distribution for land and water across the entire dataset, without temporal classification.

arise due to differences in spectral bands, response functions, spatial resolutions, viewing angles, and atmospheric correction methods [27], [28], [29]. As shown in Fig. 2, reflectance values at the water–land boundary, derived from actual water levels in four reservoirs in South Korea, exhibit significant temporal and regional variability, as illustrated by the left set of graphs. The right set of graphs in Fig. 2, which summarizes all data without temporal separation, shows substantial variability across regions. For example, the land reflectance range in Chungju overlaps with water reflectance range in Andong and Yongdam. This indicates that a universal threshold cannot be reliably applied across different satellites, regions, or times. Alternatively, recent studies have applied machine learning classifiers and deep segmentation models to surface water mapping from multispectral and SAR imagery, achieving strong accuracy but often requiring region-specific training data [30], [31].

While previous studies have attempted to monitor water levels using altimetry, optical indices, or machine learning

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classifiers, their applicability remains constrained by sparse revisit times, fixed-threshold assumptions, or dependence on region-specific training data. These limitations hinder transferability across different sensors, regions, and hydrological conditions. To address these limitations, we introduce a DEM-driven, per-scene dynamic thresholding approach that directly exploits reflectance (optical) and backscattering (SAR) properties without relying on fixed indices or pretrained models. This methodological shift enhances adaptability across both temporal and regional variations, enables direct cross-sensor integration, and supports seamless extension from optical to SAR datasets. By doing so, our study provides a scalable framework for constructing accurate, high-frequency reservoir water-level time series in data-sparse regions, particularly where conventional methods fall short.

The structure of this article is as follows. In Section II, the data and methodology outline the study area, datasets [including HLS], and the proposed methodology. In Section III, the results present the application results and examine the potential extension of this methodology to SAR satellite data, which in this study were accessed and preprocessed using Google Earth Engine (GEE). This highlights the framework's reproducibility and scalability through standardized cloud-based pipelines while also leveraging SAR's ability to overcome the limitations of optical sensors under cloudy conditions [32]. Finally, in Section IV, the discussion highlights the key findings and contributions of this research and discusses potential directions for future work.

II. Data and Methodology

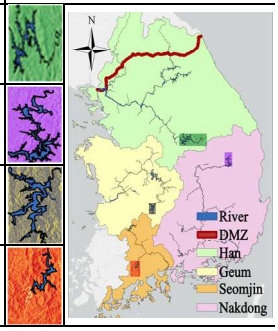
A. Study Area

Although the proposed methodology is generic and designed for global applicability, its performance is demonstrated using four reservoirs in South Korea; Korean Peninsula was chosen due to the availability of high-quality in situ data and its diverse hydrological conditions, which provide a representative and robust setting for testing the proposed approach. The peninsula encompasses both South and North Korea and includes transboundary rivers—an illustrative case of water conflicts arising from the lack of hydrological information sharing between the two nations [5], [33]. Furthermore, the study area exhibits pronounced seasonal variability, influenced significantly by the impacts of climate change. Summers are marked by high precipitation and humidity, while winters are typically dry and cold [34]. As will be discussed later, the wet summer season poses challenges in interpreting optical satellite data due to increased cloud cover, which significantly affects data availability and quality. This limitation serves as a key motivation for transitioning to the complementary use of SAR data, which is not affected by atmospheric conditions.

Table I summarizes the characteristics of the four reservoirs selected as study sites, each situated in one of South Korea's four major watersheds: Han, Geum, Nakdong, and Seomjin Rivers.

These four multipurpose reservoirs serve critical functions in their respective basins—providing flood control and water storage during the summer monsoon season while ensuring adequate domestic, agricultural, and environmental water supply during the dry season. As shown in Table I, they exhibit distinct hydrological behaviors, with operational ranges greater than 25.5 m in water level. Consequently, their water surface areas vary significantly, with maximum-to-minimum area ratios ranging from 3.0 to 5.0, reflecting the dynamic and complex morphology of each system. This complexity is visually illustrated in Fig. 5(b) and (c), consistent with prior reports for those watershed predominantly forested, rugged catchments with narrow valleys [35], [36], [37]. In addition to their hydrological complexity, these reservoirs offer near-real-time in situ data (10-min, hourly, and daily intervals), accessible via Korea Water Resources Corporation (https://www.water.or.kr/, as of June 2025). This data availability facilitates direct validation of remote sensing-derived water levels, enhancing the reliability of the proposed methodology.

TABLE I
Characteristics of the Four Reservoirs Selected as Study Sites in South Korea, Categorized by Watershed

Dam (Watershed)	Operational level range (m)	Surface area (km², Max / Min)	Comprehensive Visualization
Chungju (Han)	35.0	133.9 / 45.3	
Andong (Nakdong)	31.7	80.1 / 22.4	
Yongdam (Geum)	37.0	52.8 / 12.5	
Juam (Seomjin)	25.5	41.1 / 8.2	

B. Data

As previously mentioned, the spatial resolution of optical satellites varies significantly depending on the type of satellite, influenced by factors such as the level of detail, spatial coverage, and revisit time. It is important to note that higher resolution does not always equate to better performance. While high-resolution imagery provides detailed observations, it can also amplify shadow effects in complex terrains and requires significant computational resources, which may limit its practical applications [38], [39]. In addition, high-resolution data are often available only through commercial sources, further restricting accessibility. Therefore, selecting appropriate remote sensing data is essential, balancing the specific requirements and constraints of each application [24].

Considering these tradeoffs, this study focuses on relatively small and complex water bodies—such as the narrow and meandering example shown in Fig. 1—and aims to construct water-level time series at a finer temporal resolution. To achieve this, we utilize HLS data products, a publicly accessible form of passive optical satellite data. HLS merges

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TABLE II
CONFIGURATION OF HLS DATA

Component	Description
Agency	NASA (USGS), ESA
Mission & Launch Date	Landsat 8: Feb. 2013 / Sentinel-2A: Jun. 2015 / Landsat 9: Sep. 2021 / Sentinel-2B: Mar. 2017
Data Products	HLSL (Landsat), HLSS (Sentinel)
Multi-spectral Bands	HLSL: 9 bands HLSS: 12 bands
Revisit Time	~2–3 days (theoretical, combined)
Spatial Resolution	HLSL: 30 m HLSS: 10 m, 20 m, 60 m (original), 30 m (resampled)

imagery from Landsat-8/-9 and Sentinel-2, providing medium-resolution observations at a spatial resolution of 30 m. In addition to their adequate resolution, HLS data are freely accessible and temporally stackable, making them well suited for the objectives of this study [18]. A summary of the key specifications of the HLS data is provided in Table II.

A key advantage of the HLS dataset is its "stackability"—the use of a consistent coordinate system and projection, along with corrections for spatial and radiometric differences between Landsat and Sentinel-2. This ensures temporal and spatial consistency, removing the need for additional preprocessing when integrating multisatellite data. HLS provides a revisit time of approximately three days, significantly shorter than that of Landsat alone (16 days) or Sentinel-2 alone (five days). It includes HLSL (Landsat-8, 30-m resolution) and HLSS (Sentinel-2, native resolution up to 10 m). For this study, both datasets were used: HLSL from April 2013 and HLSS from December 2015 to December 2022. Although HLSS provides a maximum resolution of 10 m, it was resampled to 30 m for three main reasons: 1) to assess the applicability of the methodology before extending it to higher resolution data; 2) to reduce computational resource demands; and 3) to ensure spatial consistency between the two datasets, thereby maximizing the "stackability" advantage of the HLS dataset. In contrast, relying on Sentinel-2 imagery alone would restrict the temporal record to post-2015, whereas HLS extends it back to 2013 while maintaining radiometric consistency. All data were accessed via NASA's AρρEEARS (https://appeears.earthdatacloud.nasa.gov/, as of June 2025) platform.

Another important advantage of the HLS products is their provision of multispectral bands, with HLSL and HLSS offering 9 and 12 bands, respectively. Unlike MODIS, which provides band data at varying spatial resolutions, all bands in the HLS products share the same spatial resolution. This consistency allows for seamless combination of bands when calculating reflectance indices, such as the normalized difference water index (NDWI), without any loss of resolution. In this study, we employed both individual bands and band combinations to compute NDWI—a widely used water index—for comparative analysis against our proposed method, which is specifically designed to address a key limitation of single-band approaches: the challenge of establishing a robust threshold. Particular emphasis was placed on shortwave infrared 1 (SWIR1) 1.57–1.65-μm band, which has been shown to be highly effective for water detection, as originally noted by Tucker [40]. Subsequent studies have further confirmed the effectiveness of SWIR1 in distinguishing water from surrounding land surfaces. These studies consistently demonstrate strong performance across diverse environments and methodological frameworks [24], [41], [42]. Building on these established findings and data characteristics, we developed a robust methodology that leverages the spatial consistency and spectral sensitivity of HLS imagery, particularly the SWIR1 band, to extract accurate water levels through a dynamic thresholding scheme. Section II-C details this methodology, which is further extended to SAR imagery to overcome limitations inherent to optical observations.

C. Methodology

1) Identifying Water–Land Boundaries Using Maximum Reflectance Difference for Water-Level Estimation: The basic principle for extracting water levels relies on the distinct reflectance characteristics of water and land. Water typically exhibits low reflectance in the near-infrared (NIR) and SWIR bands, whereas land surfaces show higher reflectance. Building on this principle, water boundaries can be delineated and combined with DEM data to derive elevation values at the boundary, which are then used as water levels. This conventional workflow, illustrated in Fig. 3 (A0–A3), generally applies fixed thresholds, which are simple but often inaccurate under varying conditions [24].

To address these limitations, we propose a novel methodology that is effective across various time scales and particularly suitable for small and complex water bodies. As shown in Fig. 3 (B0–B3), the method begins by generating DEM-driven water masks (DWMs) (B0), as detailed in Section II-C.2 DEM-DWM. These water masks—each representing a distinct water level—are used to extract only the corresponding pixels from the imagery (B1). Then, the elevation level, at which the greatest difference in average reflectance occurs across adjacent water masks, is identified (B2). This level is interpreted as the actual water level, and the corresponding reflectance value becomes a dynamic threshold unique to that scene (B3).

The proposed approach works under the assumption that "the largest difference in average reflectance occurs at the water–land boundary, and the elevation corresponding to this boundary will be the actual water level." This logic is explained in detail in Fig. 4 wherein the average reflectance values of pixels within each water mask are plotted on the y-axis against the corresponding water mask level on the x-axis. As the water level increases (i.e., as the x-axis value rises), the slope of the curve also increases ($\partial y/\partial x > 0$) until it reaches the actual water level (marked by the red line). Beyond this point, the rate of increase diminishes ($\partial y/\partial x < 0$), forming an S-shaped pattern. This distinct pattern is a theoretically grounded characteristic of reflectance values and is

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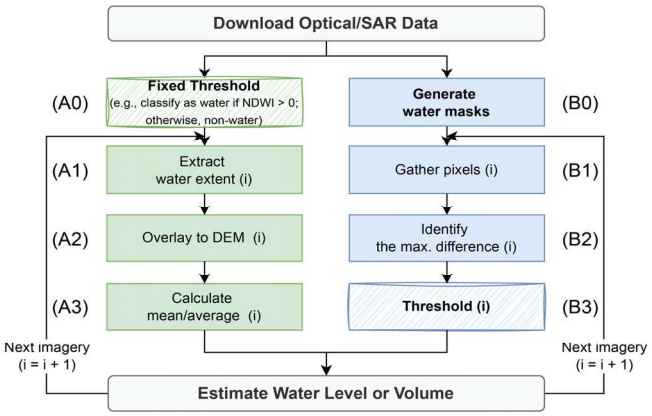


Fig. 3. Flowchart comparing the conventional method and the proposed approach for extracting water level or volume from optical and SAR imagery data. The left branches (A0–A3) represent the conventional method, which relies on extracting water extent, overlaying it onto a DEM, and computing elevation statistics. The right branches (B0–B3) illustrate the proposed approach, which generates DEM-DWMs, extracts pixel reflectance values, and identifies the maximum difference in reflectance to determine water levels.

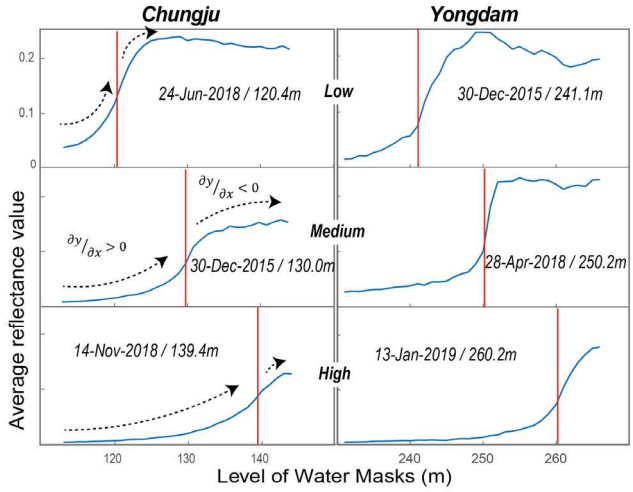


Fig. 4. Graph depicting average reflectance values and water mask levels for Chungju and Yongdam Reservoirs at low-, medium-, and high-water levels. Each panel corresponds to a specific observation date, with the actual water levels indicated. The vertical red lines mark the true water levels, demonstrating the point at which the reflectance change stabilizes. Note that, the distinct reflectance patterns occur before and after the actual water level. For spatial visualization of these water surface changes, representative maps are provided in Fig. S. 1.

true across all cases of water level—low, medium, and high. The identification of this property serves as the foundation for identifying the actual water level.

In summary, the final step of the proposed methodology identifies the elevation corresponding to the water mask that exhibits the greatest reflectance difference (B2), while the associated pixel value becomes the dynamic threshold for that imagery (B3). This is where the most significant distinction from conventional methods emerges. Rather than relying on a fixed-threshold regardless of spatial or temporal variability, the proposed approach derives a threshold adaptively for each scene. This dynamic formulation enhances transferability

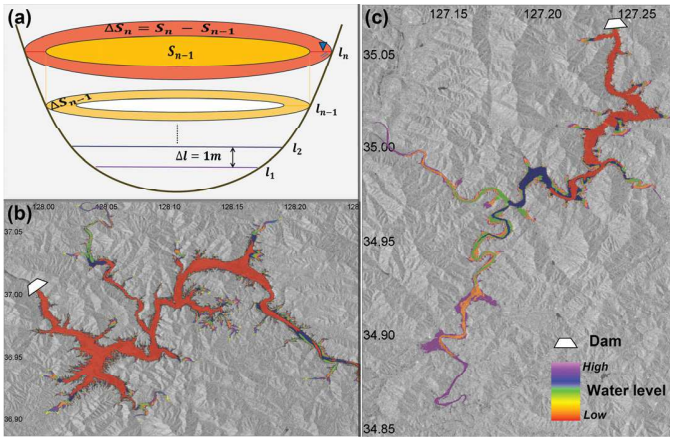


Fig. 5. Generation of DEM-DWMs at 1-m intervals for pixel selection in dataset composition. (a) Conceptual illustration of the DWM methodology, where l_n represents the water level at the nth interval, S_n denotes the water surface at the nth water level, and ΔS_n refers to the area of the nth water surface excluding the surface of the $(n-1)$th level. (b) and (c) Application of this approach to Chungju and Juam Reservoirs, with color gradients representing water-level variations from low (red) to high (purple).

across sites and sensors while maintaining conceptual simplicity and robustness.

2) DEM-DWM: The first step in the proposed methodology is the creation of DEM- DWMs, as shown in Fig. 3 (B0). As illustrated in Fig. 5(a), DWMs are generated at 1-m intervals, spanning from low to high water levels, with the water surface at each level (l_n) represented as S_n. Instead of utilizing the entire water surface at a given height, this study employs a ring-shaped water surface (ΔS_n), which excludes the water surface of the previous height (S_{n-1}). This approach enhances the sensitivity of surface characteristics at the given height, improving the accuracy of water-level estimation. Fig. 5(b) and (c) highlights the small and complex nature of the study area, characterized by intricate land–water interfaces. These features pose unique challenges for water-level extraction due to their irregular topography and fragmented water boundaries. Addressing these complexities necessitates the use of advanced methodologies, such as DWMs, to ensure accurate and reliable results. Nevertheless, the reliability of the proposed methodology is closely tied to the quality of the DEM, and high-resolution DEMs are recommended to ensure accurate water-level estimation. For validation, this study employed the LiDAR-derived DEM with 1-m spatial resolution, which was previously used in [5] to generate water masks for altimetry-based analyses, and offers substantially higher accuracy than global DEM products (e.g., SRTM, ALOS, and AW3D).

3) Application of t-Test for Water Mask Differences: To determine the elevation level corresponding to the maximum reflectance difference, one could simply apply the first-order difference (slope) or the second-order difference or curvature, as shown in the following equations:

$$\text{Slope} : \Delta y = y_{i+1} - y_i \tag{1}$$

$$\begin{aligned}\text{Curvature} : \Delta^2 y &= \left(y_{i+2} - y_{i+1}\right) - \left(y_{i+1} - y_i\right) \\ &= y_{i+2} - 2y_{i+1} + y_i\end{aligned} \tag{2}$$

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where y_i is the average reflectance value of pixels associated with the water mask at level (i).

While these gradient-based approaches generally perform well with normal data, the pixel groups forming the dataset exhibited relatively high standard deviation. This indicates the presence of significant noise and a higher likelihood of errors. In addition, as observed in Fig. 4, beyond the actual water level (red line), the slope tends to decrease and become irregular at higher water levels due to the diverse and mixed surface types such as sand and grass, leading to substantial errors in the results. To address these challenges, this study employed the t-test to determine the significance of differences between two groups. Instead of interpreting the p-value, the t-value, as expressed in (3), was applied to assess the magnitude of the difference as a statistical measure [43]

$$\text{t-value} : t = \frac{y_{i+1} - y_i}{\sqrt{\frac{s_{i+1}^2}{n_{i+1}} + \frac{s_i^2}{n_i}}} \tag{3}$$

where y_i represents the mean value of pixels associated with the water mask at level i, s_i denotes the sample variance of the pixels, and n_i is the number of pixels.

The t-test was not used for hypothesis testing but to quantify the separation between adjacent water mask levels, with the t-value indicating effect size. Given that each mask contains roughly 1000 pixels, the central limit theorem supports the approximate normality of sample means, validating the t-test in this context [44]. To enhance data quality, distorted areas identified as high aerosol, snow/ice, cloud shadow, adjacent to cloud/shadow, or cloud were detected and removed using the improved mask (Fmask) algorithm proposed by Zhu et al. [45]. The t-value was then calculated using the filtered dataset to identify the elevation level where the spectral separation between land and water was maximized. A detailed step-by-step description of the proposed method is provided in Appendix A of the Supplementary Materials.

4) Evaluation Metrics: To assess the reliability and accuracy of the proposed methodology, several statistical indices were employed. Specifically, the unbiased root-mean-squared error (ubRMSE) and the coefficient of determination (R^2) were used as quantitative indicators. These metrics were selected to evaluate the magnitude of agreement between water levels extracted from satellite data and those measured through ground-based observations.

The ubRMSE metric, defined in (4), quantifies the error by eliminating bias that could skew the results, providing a robust measure of accuracy

$$\text{ubRMSE} = \sqrt{\mathbb{E}\left[(l_{\text{sat}} - l_{\text{real}})^2\right] - b^2} \tag{4}$$

where l_{real} is the actual measured water level (ground truth), l_{sat} is the estimated water level (from satellite), and b is the mean bias defined as $b = \mathbb{E}[l_{\text{sat}}] - \mathbb{E}[l_{\text{real}}]$.

The coefficient of determination (R^2), shown in (5), indicates the proportion of variance in the measured water levels (l_{real}), that is, predictable from the satellite-derived water levels (l_{sat})

$$R^2 = 1 - \frac{\sum_{i=1}^{n} \left(l_{\text{real},i} - l_{\text{sat},i}\right)^2}{\sum_{i=1}^{n} \left(l_{\text{real},i} - \bar{l_{\text{real}}}\right)^2} \tag{5}$$

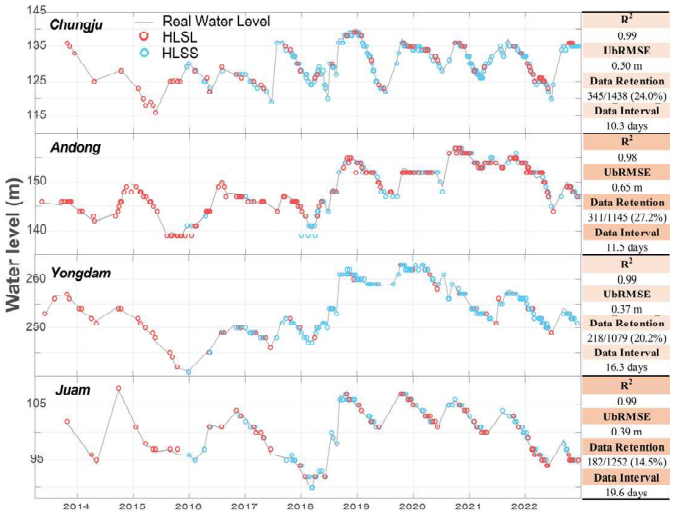


Fig. 6. Water-level time series extracted from HLS for each reservoir, comparing satellite-derived estimates (HLSL and HLSS) with actual measured water levels. The data retention rate and actual observation interval are indicated for each site, highlighting the impact of cloud contamination on data availability. Statistical metrics, including R^2 and ubRMSE, demonstrate the accuracy of the retrievals across different hydrological conditions.

where $\bar{l_{\text{real}}}$ is the mean of the measured water levels.

In addition to ubRMSE and R^2, mean bias was also assessed to provide further insight into the accuracy of the retrievals. It may be noted that there are several other classification metrics, such as overall accuracy (OA, the proportion of correctly classified instances) or the $F1$-score (the harmonic mean of precision and recall), which are also available in the literature and could have been applied. However, we believe that the chosen metrics offer a comprehensive evaluation of satellite-based water-level estimates by capturing both the magnitude of overall error and the statistical agreement with in situ observations. By incorporating these robust evaluation metrics, the study ensures a rigorous and precise validation of the methodology, demonstrating its effectiveness in accurately capturing water-level dynamics using satellite data.

III. Results

A. Application of HLS (Optical) Data

Using the DEM-DWMs and two types of HLS datasets—HLSL (2013, April–2022, 9.8 years) and HLSS (2015, December–2022, 7.1 years)—with the SWIR1 band as the primary data source, the proposed methodology successfully extracted water levels. As shown in Fig. 6, the results closely matched the actual measured water levels. The reservoirs analyzed, located in hydrologically distinct and geographically distant watersheds, demonstrated high accuracy in water-level estimation. For the four target reservoirs, the coefficient of determination (R^2) values were near-perfect, approaching 1, indicating an excellent model fit between the extracted and observed water levels. Moreover, the unbiased root-mean-squared error (ubRMSE) ranged from 37 to 65 cm, further validating the high precision of the proposed approach.

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Despite these promising results, data availability emerged as a persistent challenge due to the inherent limitations of optical satellite data, particularly its susceptibility to cloud interference. This issue is exacerbated by the geographical and meteorological characteristics of Korean Peninsula. While the mountainous terrain facilitates reservoir construction, the summer monsoon season, characterized by concentrated rainfall during a short period, considerably increases cloud cover. After applying the Fmask algorithm to detect and remove distorted data, the number of usable datasets was greatly reduced. For instance, at Andong Reservoir, only 27.2% of the datasets remained valid after filtering, resulting in an effective observation interval of 11.5 days. The issue was even more pronounced for Juam Reservoir, where just 14.5% of the datasets (182 out of 1252) were usable over the ten-year study period, yielding a sparse average revisit time of 19.6 days.

B. Utilization of a Single Band

While earlier studies reported successful use of single infrared bands for water mapping [41], more recent research has largely shifted toward multiband indices. This shift reflects the difficulty of establishing a single, robust threshold that works consistently across different times and locations [24]. On the other hand, the wide variety of indices, including NDWI, modified NDWI (MNDWI), and automated water extraction index (AWEI), further adds complexity, as no universally accepted standard exists for selecting the most appropriate index [46].

Consequently, researchers or users must choose based on specific study areas, goals, and data conditions. Acknowledging these limitations, this study focused on extracting water levels using only the SWIR1 band. To further explore the potential of single-band applications, we also calculated the widely used NDWI, as defined by McFeeters [28], using (6), and compared its performance with individual bands, Band 3 (green) and Band 8 (NIR), which compose NDWI, together with Band 11 (SWIR1). This comparison was designed to test whether single-band applications can achieve performance comparable to or better than NDWI, highlighting the effectiveness of the proposed methodology

$$\text{NDWI} = \frac{\text{Green} - \text{NIR}}{\text{Green} + \text{NIR}} \tag{6}$$

where green (Band 3): 0.53–0.59 μm and NIR (Band 8): 0.78–0.88 μm.

For this analysis, we used data from HLSS's 12 bands across four reservoirs over the entire study period. The comparative results are presented in Fig. 7. In our proposed dynamic threshold approach, we investigated the utility of NDWI as a surrogate for reflectance, alongside reflectance from different retrieved bands. The results indicated that datasets corresponding to Band 11 (SWIR1) consistently achieved the highest accuracy, with R^2 values of 0.99 for all reservoirs except Andong (0.97), and ubRMSE values no greater than 0.74 m across all sites, demonstrating stable and outstanding performance across diverse conditions. In contrast, NDWI showed much larger spatial variability. While Yongdam achieved an

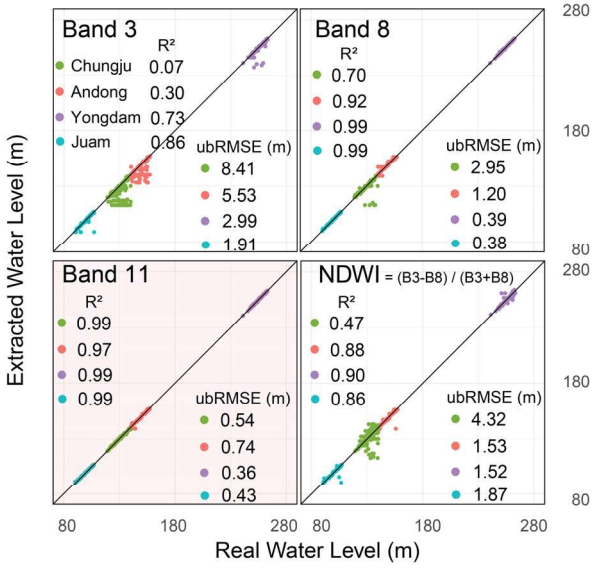


Fig. 7. Comparison of water-level retrieval accuracy using individual spectral bands (Bands 3, 8, and 11) and the NDWI index from HLSS over the full study period (2015–2022). The results demonstrate that SWIR1 (Band 11) achieves the highest accuracy, surpassing NDWI, which relies on multiple bands. For reference, the results obtained using the conventional fixed-threshold NDWI method are provided in Fig. S. 2.

R^2 of 0.90, Chungju exhibited a markedly lower R^2 of 0.47, and ubRMSE reached up to 4.32 m, indicating strong site-to-site inconsistencies. When NDWI was applied conventionally with a fixed threshold (Fig. S. 2), the results were even poorer, with R^2 values of only 0.47–0.71 and ubRMSE of 2.61–4.02 m. These findings highlight that the use of individual bands—particularly Band 11 (SWIR1) and Band 8 (NIR)—offers more reliable and superior performance than relying on NDWI.

This can be attributed to the fact that multiband indices must operate at the coarsest spatial resolution among the input bands—a particular drawback for sensors like MODIS, where individual bands vary in resolution. As a result, their effectiveness in capturing fine-scale water surface dynamics is compromised. In contrast, using a single band allows exploitation of the sensor's highest native resolution, enabling more precise delineation of water boundaries. Although indices such as NDWI are widely used, their dependence on multiple spectral bands inherently limits spatial resolution and consequently the accuracy of surface water detection.

Our results demonstrate that the SWIR1 band alone is sufficient for accurate water identification in certain contexts, eliminating the need for multiband indices. This simplification enhances applicability across diverse satellite platforms, including SAR systems, which lack multispectral capabilities but possess a distinct advantage: the ability to overcome the limitations of optical satellites under cloud-covered conditions, as discussed in Section III-A.

C. Utilization of SAR (Sentinel-1) Data

To address the cloud coverage limitations inherent to optical satellites, a common approach is to integrate data with SAR

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TABLE III
CONFIGURATION OF SENTINEL-1 (SAR) DATA

Component	Description
Operating Agency	European Space Agency (ESA)
Mission	Earth Observation (Synthetic Aperture Radar, SAR)
Satellite Series Launch Date	Sentinel-1A: April 3, 2014 Sentinel-1B: April 25, 2016 – Decommissioned in 2021 Sentinel-1C: Planned
Revisit Time	~12 days (single satellite) ~6 days (combined operation of Sentinel-1A & 1B)
Polarimetry	Single/Dual Polarimetry (VV, VH, HH, HV)
Frequency Band	C-band (5.405 GHz)
Observation Mode	IW (Interferometric Wide Swath) EW (Extra-Wide Swath) SM (Strip Map)
Swath Width	~250 km (IW mode)
Spatial Resolution	~5m × 20m (IW mode)



Fig. 8. Water-level time series extracted from Sentinel-1 (SAR) for each reservoir, comparing satellite-derived estimates with actual measured water levels. The data retention rate and actual observation interval are indicated for each site, demonstrating the advantage of SAR in maintaining continuous observations unaffected by cloud contamination. Statistical metrics, including R^2 and ubRMSE, highlight the accuracy of the retrievals across diverse hydrological conditions.

imagery, which operates independently of cloud interference. However, due to the differing revisit times of optical and SAR satellites, it is often challenging to obtain temporally concurrent images. Instead of solely using SAR imagery to supplement optical data, this study aimed to independently construct water dynamics by directly applying the proposed methodology to backscattering data from Sentinel-1, a representative SAR satellite.

This approach is feasible for two main reasons. First, as previously noted, cloud cover significantly reduces the availability of optical imagery, making SAR data a valuable alternative. Second, the methodology proved sufficiently accurate when applied to a single band, suggesting its applicability to SAR data, which offers fewer spectral options. As summarized in Table III, SAR satellites typically lack the multispectral capabilities of optical sensors but instead provide polarization modes—such as horizontal–horizontal (HH), vertical–vertical (VV), horizontal–vertical (HV), and vertical–horizontal (VH)—that reflect distinct microwave scattering properties. In this study, the VV polarization mode was selected as the primary dataset due to its proven effectiveness in water detection, aligning with the use of the SWIR1 band in the optical domain. This choice is further supported by recent studies, such as Tian et al. [47], which highlight its reliability for water detection under diverse environmental conditions.

The original resolution of Sentinel-1 SAR data is 5 × 20 m. To ensure consistency with the resolution of the previously utilized HLS (optical) data, the Sentinel-1 data were resampled to a 30-m resolution using GEE [48]. This adjustment not only aligned the datasets but also struck a balance between computational efficiency and data accuracy, as processing higher resolution data demands significantly more resources without substantial benefits for the study's objectives. Furthermore, due to a power supply issue, Sentinel-1B ceased operations on December 23, 2021. Consequently, data collected after this date relied exclusively on Sentinel-1A imagery for analysis [49].

As shown in Fig. 8, one of the key advantages of SAR data is its immunity to cloud interference, allowing the use of the entire dataset without filtering. This characteristic eliminates the limitations faced by optical data, where cloud coverage significantly reduces data availability. In terms of accuracy, the application of the proposed methodology to SAR data yielded impressive results, R^2 values $\geq$ 0.95 and ubRMSE values ranging from 0.47 m (Juam) to 1.29 m (Chungju), demonstrating strong model performance under varying hydrological conditions. The relatively larger ubRMSE values in Chungju and Andong compared to Yongdam and Juam can be explained by the greater shoreline complexity of these reservoirs, as they cover substantially larger areas (Table I), which increases the likelihood of including a few large outliers. For example, Fig. S. 3 illustrates two observations at Chungju Reservoir with nearly identical water levels. On January 21, 2018, the retrieval was accurate, but on February 2, 2018, a significant error occurred because speckle overlapped the reservoir extremities. This case demonstrates how boundary complexity and speckle noise can amplify local errors, contributing to higher ubRMSE in larger reservoirs.

Despite its advantages, SAR data presented challenges due to geometric distortions such as foreshortening, layover, and shadowing from terrain variations, as well as speckle noise that can increase data variability [50], [51]. In this study, we relied on the standard Sentinel-1 GRD products without applying additional preprocessing. This was intentional; kernel-based speckle filtering could blur the sharp water–land boundary

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TABLE IV

ACCURACY AND AVAILABILITY OF EXTRACTED WATER LEVELS FOR CHUNGJU RESERVOIR. METRICS INCLUDE UBRMSE, MEAN BIAS, AND R^2. THE MEAN ANNUAL OBSERVATION COUNT REPRESENTS THE AVERAGE NUMBER OF VALID OBSERVATIONS PER YEAR (2020–2022), AND THE MEAN OBSERVATION INTERVAL ACCOUNTS FOR ACTUAL DATA AVAILABILITY

No.	Satellite/ Dataset	Type	Year of Launch	***ubRMSE*** (m)	Mean Bias (m)	R^2	Mean Annual Observation Count	Mean Observation Interval (days)
1	IceSAT2	Altimetry	2018	0.28	0.12	>0.99	6.5	56.2
2	Sentinel-3A		2016	0.19	-0.07	>0.99	11.2	32.6
3	Sentinel-3B		2018	0.34	-0.26	0.99	13.3	27.4
4	Sentinel-1A&B	SAR	2014	1.29	-0.40	0.95	39.9	9.1
5	HLSL	Optical	2013	0.40	-0.14	>0.99	7.3	50.0
6	HLSS		2015	0.54	-0.27	0.99	39.1	9.3
7	CryoSat2	Altimetry	2010	1.40	0.25	0.95	6.7	54.5
All satellites/ Datasets		-	--	0.96	-0.23	0.97	124.0	2.94

that is central to our methodology, while terrain corrections may introduce artifacts that vary with acquisition geometry. By constraining the analysis within the maximum water extent and leveraging large sample sizes per mask, our approach mitigates much of this variability while retaining boundary precision. Nevertheless, residual speckle and geometric effects remain and may still influence backscattering values, which partly explains the variability observed in our results.

D. Comprehensive Water-Level Time Series Through Integration of Multiple Satellite Data

This study was motivated by the limited temporal resolution of altimetry data, which, despite providing highly accurate water-level measurements, offers only a few observations per year, restricting its practical application for frequent monitoring, particularly in small and complex water bodies [5]. To address this limitation, we employed an optical–SAR integration approach to supplement altimetry-based observations, significantly enhancing temporal resolution. Optical (HLS) and SAR (Sentinel-1) datasets offer much more frequent observations, and by integrating these sources, we constructed a denser water-level time series that achieved a strong model fit, marking a substantial improvement over using only altimetry data.

Table IV summarizes the accuracy and availability of each dataset for Chungju Reservoir, the only reservoir in the study traversed by all four altimetry satellites [5]. The table rank results in descending order of accuracy, highlighting the tradeoffs between temporal availability and retrieval accuracy. The results demonstrate that altimetry satellites provided the highest accuracy, with ubRMSE values ranging from 0.19 to 0.34 m (except for CryoSat-2, which had higher errors of 1.40 m). However, their low temporal availability (6.5–13.3 observations per year) limits their suitability for frequent monitoring. In contrast, optical (HLS) and SAR (Sentinel-1) satellites, while slightly less accurate, provided significantly greater data availability, with 46.4 and 39.9 valid observations per year, respectively.

By integrating all available satellite datasets, we achieved an average observation interval of just 2.94 days, offering a substantial improvement over altimetry-only methods. These findings highlight the tradeoffs between accuracy and observation frequency, reinforcing the value of multisatellite fusion in enhancing global water monitoring capabilities.

IV. DISCUSSION

While the integration of multisatellite datasets demonstrated considerable improvements in both temporal resolution and spatial coverage, certain limitations remain.

Most notably, the accuracy of both optical and SAR data was affected by inherent challenges—cloud interference in the case of optical satellites, and speckle noise and geometric distortions (e.g., foreshortening, layover, and shadowing) for SAR. Although several filtering techniques (e.g., Fmask) have already been developed, addressing these issues will require further improvements in preprocessing techniques, including advanced atmospheric correction, noise suppression, and terrain normalization.

In addition, while the performance of the proposed methodology depends on the accuracy of optical and SAR data, it equally, if not more, depends on the quality and resolution of the DEM, which serves as the foundation for generating water masks. In regions where high-quality DEMs are unavailable, such as transboundary areas with limited access, the overall accuracy of water-level estimation may be significantly compromised. In such cases, commercial services that generate bathymetric data using satellite-derived elevation models may be considered. Alternatively, open-source satellite-derived bathymetry [20], [21] offers a viable option, although these datasets are often characterized by lower spatial resolution and complex processing workflows, which can still impact the accuracy of results. Furthermore, as the structure and processing methods of such datasets differ across studies,

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reproducing them often demands considerable time and computational resources.

In addition, the reliance on existing open-access datasets limits the spatial and temporal resolution that could otherwise be achieved using commercial satellite platforms, which typically offer more frequent revisit cycles and finer spatial detail. However, a key strength of the proposed methodology lies in its foundation on the physics of optical reflectance and SAR backscattering. As the proposed approach is independent of the sensor-specific characteristics, it can be broadly applied to a wide range of open-access SAR and optical satellites, including RISAT-1 and RISAT-2, ALOS-2, and RADARSAT-2 (SAR), as well as Sentinel-2, SPOT, PlanetScope, and ASTER (optical).

Looking ahead, future research should focus on the following conditions.

1) Scaling to global applications by integrating multisensor observations across platforms and applying it to diverse hydrological contexts and reservoir types, thereby enhancing both spatial/temporal granularity and transferability of the approach.

2) Simplifying the process of integrating satellite-derived bathymetry with DEMs, as existing methods (e.g., ICESat-2, SWOT, and Sentinel-2) are often complex and resource-intensive. Streamlined workflows would make the methodology more practical for transboundary and data-sparse regions.

3) Adapting the methodology into a near-real-time monitoring framework. While cloud-based platforms such as GEE have limitations in handling long-term archives, they are well suited for processing recently available datasets, enabling timely reservoir monitoring and supporting operational water management and hazard preparedness.

4) Designing advanced preprocessing methods that reduce speckle noise and topographic distortions without degrading the original satellite data, for example, by leveraging machine learning-based or physics-based normalization, to further improve the robustness of SAR-based retrievals.

By addressing these challenges and leveraging the rapidly growing availability of satellite data, future studies can greatly enhance the spatiotemporal resolution, consistency, and scalability of inland water monitoring—contributing meaningfully to global water security and sustainable resource management.

In terms of computational efficiency, the proposed methodology is not resource-intensive. The DEM-DWMs are generated only once and subsequently reused, and the retrieval relies on single-band reflectance or backscattering, which is computationally simpler than multiband index approaches. All analyses in this study were performed in MATLAB on a standard workstation (Intel i7 CPU, 16-GB RAM), and complete multiyear datasets for each reservoir were processed within 4–6 h. These results indicate that the method is readily applicable in data-sparse regions without the need for high-performance computing resources.

V. Conclusion

This study proposed a novel and scalable methodology for extracting water levels from satellite imagery, designed to address challenges in monitoring water dynamics of small and complex inland water bodies. By integrating multiple satellite datasets and employing DEM-DWMs with dynamic threshold adjustments, the methodology achieved high accuracy and robust applicability with both optical (HLS) and SAR (Sentinel-1) datasets. Its effectiveness was validated across diverse reservoirs in South Korea, showcasing reliability under conditions such as cloud contamination in optical data and geometric distortions in SAR while eliminating reliance on traditional indices such as NDWI.

The integration of multiple satellite datasets—altimetry, optical, and SAR—enabled the construction of a dense water-level time series with an average interval of 2.94 days. This demonstrates the practical value of multisatellite integration, balancing the exceptional accuracy of altimetry with the higher availability of optical and SAR observations.

Because the methodology is grounded in the physical principles of optical reflectance and SAR backscattering, rather than sensor-specific calibrations, it is broadly transferable to various open-access platforms such as Sentinel-2, ALOS-2, and RISAT. However, a high-quality DEM remains a critical component of this study. In the absence of reliable DEMs—particularly in inaccessible regions such as transboundary areas—alternative approaches are required, as discussed in Section IV. This highlights both the potential and the practical limitations of applying the methodology globally.

In conclusion, this methodology provides a robust framework for constructing reliable water-level time series that can support flood risk management, equitable allocation, and sustainable water use. With the growing availability of open-access satellite data and advances in remote sensing, the approach holds strong potential for global applications and for addressing the pressing challenges of water security under climate variability.

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